

Justin Lang

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EDUCATION

University of California, San Diego

M.S. in Computer Science

San Diego, CA

Sep 2026 – Jun 2028 (expected)

Coursework in progress: Algorithms, Recommender Systems, Unsupervised Learning

University of California, Santa Barbara

B.S. in Computer Science, Minor in Statistics & Applied Probability

Santa Barbara, CA

Jun 2025

GPA: 3.94 · High Honors · Distinction in the Major · College of Engineering Dean's Honors (6×)

Coursework: Databases, Algorithms & Data Structures, Computer Architecture, Machine Learning, Deep Learning

EXPERIENCE

NICE Actimize

Software Engineer, R&D — Agentic AI for Fraud Detection

Santa Clara, CA

Oct 2025 – Jul 2026

- Co-designed and later solely owned a report engine that turns a single SQL query into a deployable agent with no code changes; the team built ~50 agents on it for testing and data exploration, 5 of which shipped to production.
- Solely owned two more production components: a heavily used agent flagging first-time and anomalous customer activity, and a configuration system that tunes each agent's behavior in production with no code change or redeploy.
- Designed from scratch the SQL schema storing agent configurations and report definitions, planning for schema evolution across future agent generations while integrating with existing legacy databases.
- Wrote Helm charts and set resource limits for agent services on Kubernetes, managing the per-agent service lifecycle and documenting the deployment path and table DDL so the team could stand up new agents independently.

Huupé

Software Engineer Intern

San Diego, CA

Summers 2023, 2024

- Architected a distributed WebSocket messaging layer on Redis Pub/Sub that replaced a manual messaging process and broadcast updates across multiple Node.js API instances, unblocking a new product launch.
- Built the backend API for a smart basketball hoop and refactored 50+ blocking calls to async, improving responsiveness.

PROJECTS

Avalution — On-Disk Merkle Trie Storage Engine · UCSB Capstone with Ava Labs

Go · Fall 2024 – Mar 2025

- Built a storage engine, on a 5-person team, that stores authenticated blockchain state as a Merkle trie directly on disk, eliminating the key-value layer conventional designs stack underneath; benchmarked 40% faster than LevelDB on an 80/20 read/write workload.
- Owned the disk manager, the revision manager for versioned trie state, and a power-of-two free-list allocator that reclaims and reuses freed disk space.
- Designed a differential layer tracking in-memory mutations against persisted state, giving fast reads and edits of uncommitted revisions.

Cetacean Distribution Modeling — with Scripps Institution of Oceanography (UCSD)

R · Fall 2024 – Spring 2025

- Built spatio-temporal generalized linear mixed models predicting marine mammal density off Southern California by fusing acoustic detections, visual surveys, and eDNA observations.
- Integrated 10+ years of CalCOFI and CASE-STSE environmental data into a single species distribution model; received the Best Data Methods Award.

SKILLS

Languages: Go, C++, Python, SQL, Java, TypeScript, JavaScript, R · **Spoken:** English, Mandarin

Databases: SQL schema design, Oracle SQL, JDBC, Redis (Pub/Sub), on-disk storage engines

Systems & Infra: Kubernetes, Helm, ingress, REST APIs, WebSockets, Node.js, AWS, Google Cloud

AI / ML: Agentic system design, LLM agents, sentiment analysis, statistical modeling (GLMMs)